Capstone — Form to AI

In groups, build one end-to-end data product in 3 hours: a **Google Form** (application / enrollment) with an **image-document upload** → stored in a **Google Sheet** → **analyzed & visualized** in Colab → your **Teachable Machine** model **identifies the uploaded image** and reports its **confidence**.

Part A — Build the form (Google Forms)

1. Go to **forms.google.com** → new form. Title it (e.g. *Scholarship Application* or *Enrollment Form*).
2. Add a few simple questions (Name, Age/Grade, Course/Program, a choice question or two).
3. Add a **File upload** question titled e.g. `Upload your document` — limit to **1 file**, type **Image**.
4. **Responses** tab → green Sheets icon → **Create spreadsheet** (this links a Sheet).
5. Fill the form a few times, each time uploading an image (a photo of a document, ID, etc.).

> **Important:** a file-upload question makes responders **sign in with Google**, and uploads save to the **form owner's Drive**. In Colab, log in with that **same owner account** so the notebook can read the images.

Part B — Get your Sheet ID + image column

- Open the linked Sheet. The **Sheet ID** is the long code in the URL between `/d/` and `/edit`.
- Note the exact **title of your upload question** as it appears as a column header (e.g. `Upload your document`).

Part C — Analyze & classify (Colab notebook)

Open the **Capstone notebook** and run it top to bottom. You only edit one config cell:

```
SHEET_ID  = 'PASTE_YOUR_SHEET_ID_HERE'
IMAGE_COL = 'Upload your document'   # your upload question's exact title
```

1. **Steps 1–4:** connect to Google, load the responses into a table.
2. **Steps 5–6:** describe + **visualize** the answers (auto bar/line charts).
3. **Interpret:** write 2–3 sentences on what the charts show.
4. **Steps 7–8:** upload your **Teachable Machine** model `.zip` — it auto-arranges and loads (no folders).
5. **Step 9:** the notebook downloads each uploaded image from its Drive link and **identifies it**.

6. **Step 10: test confidence** — distribution, average, and which uploads are below 80% (need review).

🧩 The twist — unscramble the notebook first

The notebook you're given has its cells **out of order**. Before anything runs, your group must work out the **correct sequence** and rearrange the cells (use the ↑ **/** ↓ **move-cell buttons** in Colab), then run top to bottom.

Reason from **dependencies** — what has to happen before what:

- Install packages **before** importing them; import libraries **before** using them.
- Sign in to Google **before** opening the Sheet.
- Load the data **before** describing / charting it.
- Upload & load the model **before** classifying; classify **before** testing confidence.

Each cell is labelled **A–J** with a one-line hint. When the order is right, the whole thing runs with no errors.

3-hour timetable

Time	What happens
0:00–0:15	Brief, groups, pick form type + model classes
0:15–1:00	Build the form (with image upload) + collect a few responses
1:00–1:45	Connect notebook → analyze + visualize + interpret
1:45–2:30	Teachable Machine: classify the uploaded images + test confidence
2:30–3:00	Polish (findings + 1 recommendation) + dry-run the 3-min story
Hour 4	Presentations & closing

Save time on the model: reuse a Teachable Machine model from Day 2 (just classes like *valid / not valid*, *ID / not ID*) instead of training a fresh one.

Deliverables (tick to pass)

☐ Google Form with an image upload + linked Sheet ☐ Notebook connected to the Sheet
☐ Charts + written interpretation ☐ Each uploaded image identified by the model
☐ Confidence tested (average + items <80%) ☐ 3-minute story

Scoring rubric (20 points)

Criteria	Points
Form + data — working form with image upload, responses in a Sheet	4
Analysis & visualization — notebook connected, clear charts	4
Interpretation — sensible reading of the data	4
AI identification — model classifies the uploaded images	4
Confidence + presentation — confidence tested; clear 3-min story ending in a decision	4
Total	**20**

Facilitator answer key 🔑

Correct cell order for the scrambled notebook:

```
D → H → F → I → B → A → J → E → C → G
```

Order	Cell	What it does
1	D	Install packages
2	H	Import libraries
3	F	Sign in to Google (Sheets + Drive)
4	I	Open the Google Sheet into a table
5	B	Describe the data
6	A	Visualize the responses
7	J	Upload & unzip the model
8	E	Load model + predict function
9	C	Classify each uploaded image
10	G	Test the confidence

> Accept small variations that still run (e.g. B and A can swap; D+H can be one step) — the must-hold rule is: install/import → sign in → load sheet → analyze → upload/load model → classify → confidence.